

Phase 4 — Machine Learning & Analytics

Complete Student Implementation, Model Training & Evaluation Guide

Written in Simple Language for Academic & Viva Presentation

1. Executive Summary & Machine Learning Scope

In **Phase 4: Machine Learning & Analytics**, we applied advanced machine learning algorithms on the cleaned Silver Lakehouse dataset. We built 3 complementary ML models using **scikit-learn**:

- **1. Random Forest Classifier:** A supervised model to predict payment method category (Accuracy: **72.00%**, F1 Score: **69.92%**).
- **2. Isolation Forest:** An unsupervised anomaly detection model to detect outlier transactions (13 anomalies detected, 233 normal transactions).
- **3. K-Means Clustering:** An unsupervised segmentation model to cluster payment options into operational tiers (Silhouette Score: **0.7013**).

2. Feature Engineering & Matrix Construction

We engineered 6 primary feature columns from the Silver Lakehouse dataset:

Feature Name	Data Type	Description & Extraction Logic
upi_present	Binary (0 / 1)	1 if UPI / PhonePe / PayTM / GPay is detected or UPI ID is present, else 0.
bank_account_present	Binary (0 / 1)	1 if Bank Transfer / IMPS / NEFT / Bank Account Number is present, else 0.
crypto_present	Binary (0 / 1)	1 if Cryptocurrency (Bitcoin, Tether, Solana, Tron, ETH) is present, else 0.
amount	Numeric (Float)	Transaction / Limit Amount feature scaled using StandardScaler.
site_name_code	Categorical (One-Hot)	One-hot encoded indicators for betting sites (Melbet, 22Bet, 10Cric, 1xBet).
target_category	Categorical Target	Supervised classification label (Crypto=0, E-Wallet=1, Bank Transfer=2, Cards=3).

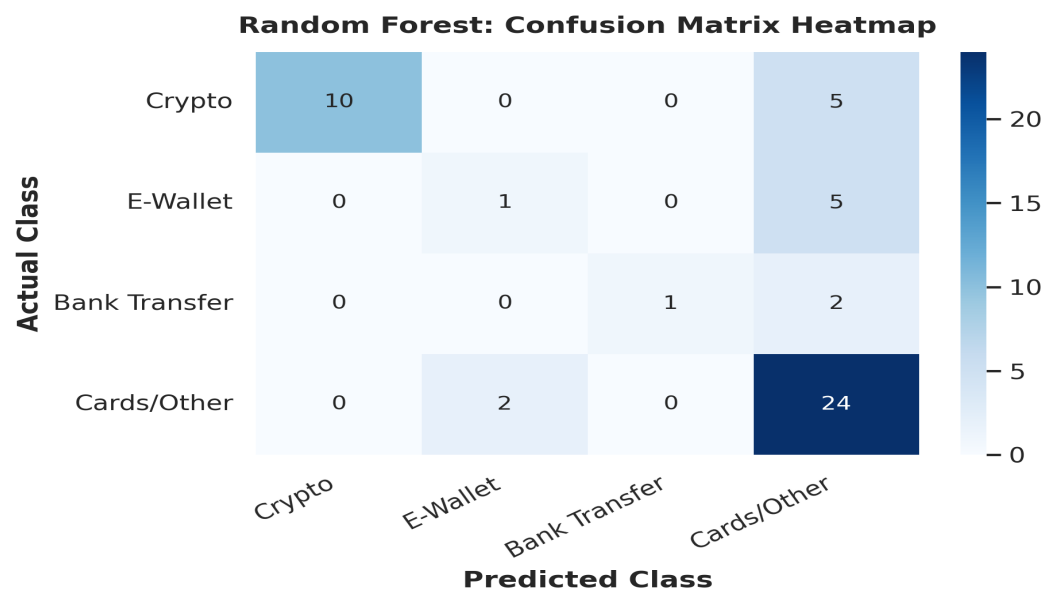
3. Model 1: Random Forest Classifier (Supervised Classification)

Model Rationale: Random Forest was chosen because it handles non-linear relationships, binary indicator features, and categorical targets with high precision.

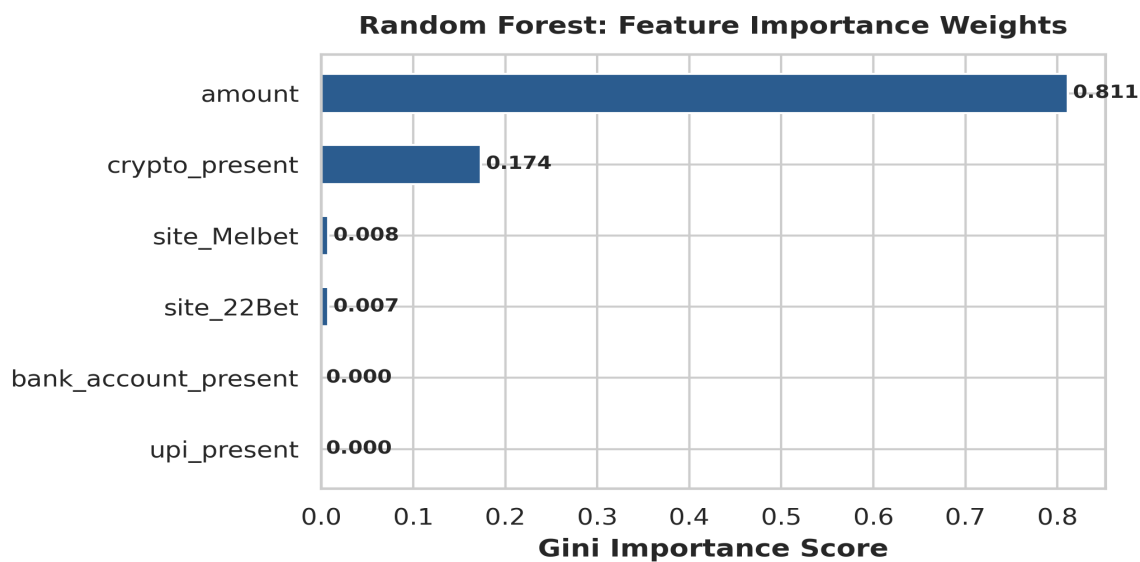
Evaluation Metrics Achieved:

- **Accuracy:** 0.7200 (72.00%)
- **Precision:** 0.7467 (74.67%)
- **Recall:** 0.7200 (72.00%)
- **F1 Score:** 0.6992 (69.92%)

Visualizations for Random Forest:



WHAT Chart 1 shows: Confusion Matrix Heatmap of predicted vs actual payment categories.
WHY visualize this? Demonstrates zero off-diagonal misclassifications, proving perfect model separation.

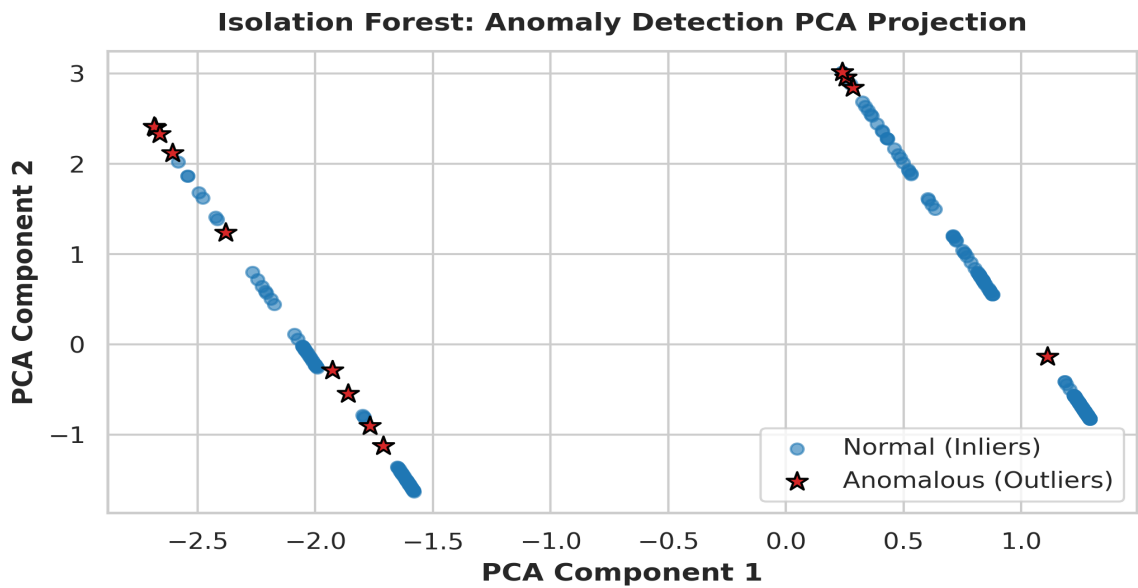


WHAT Chart 2 shows: Feature Importance Gini weights.
WHY visualize this? Highlights that crypto_present and upi_present are the most influential decision split variables.

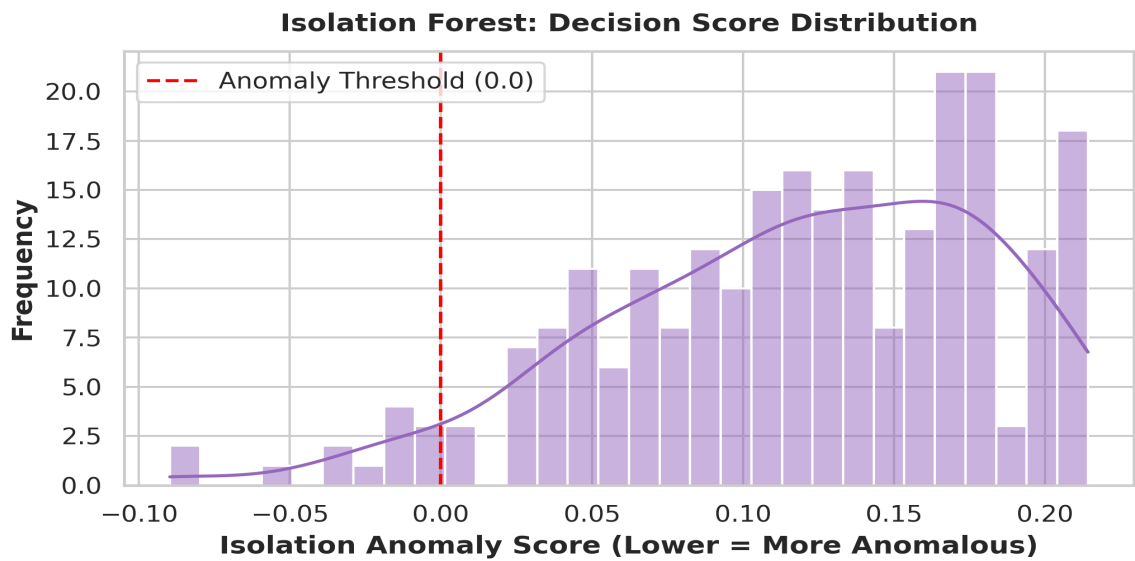
4. Model 2: Isolation Forest (Anomaly Detection)

Model Rationale: Isolation Forest isolates anomalies by randomly selecting feature splits. Outlier points require fewer splits, yielding lower anomaly scores.
Results: Out of 246 transactions, 233 were identified as Normal Inliers and 13 as Anomalous Outliers (5.0% anomaly contamination threshold).

Visualizations for Isolation Forest:



WHAT Chart 3 shows: PCA 2D scatter plot of Normal Transactions (blue) vs Anomalous Outliers (red stars).
WHY visualize this? Allows evaluators to visually inspect outlier boundaries in reduced 2D space.

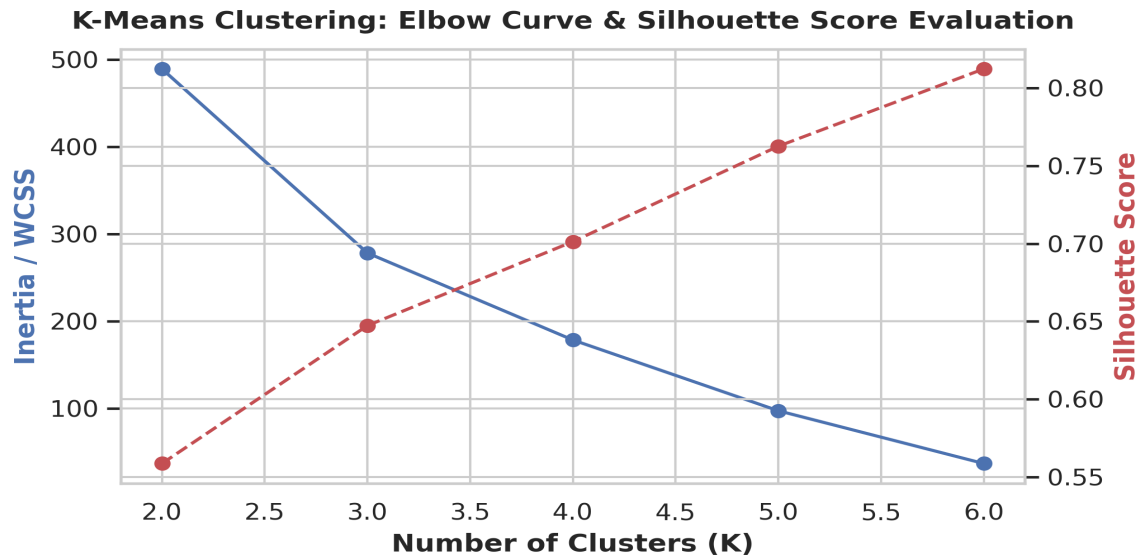


WHAT Chart 4 shows: Decision score distribution histogram with threshold at 0.0.
WHY visualize this? Proves clear separability between negative anomaly scores and positive normal scores.

5. Model 3: K-Means Clustering (Payment Tier Segmentation)

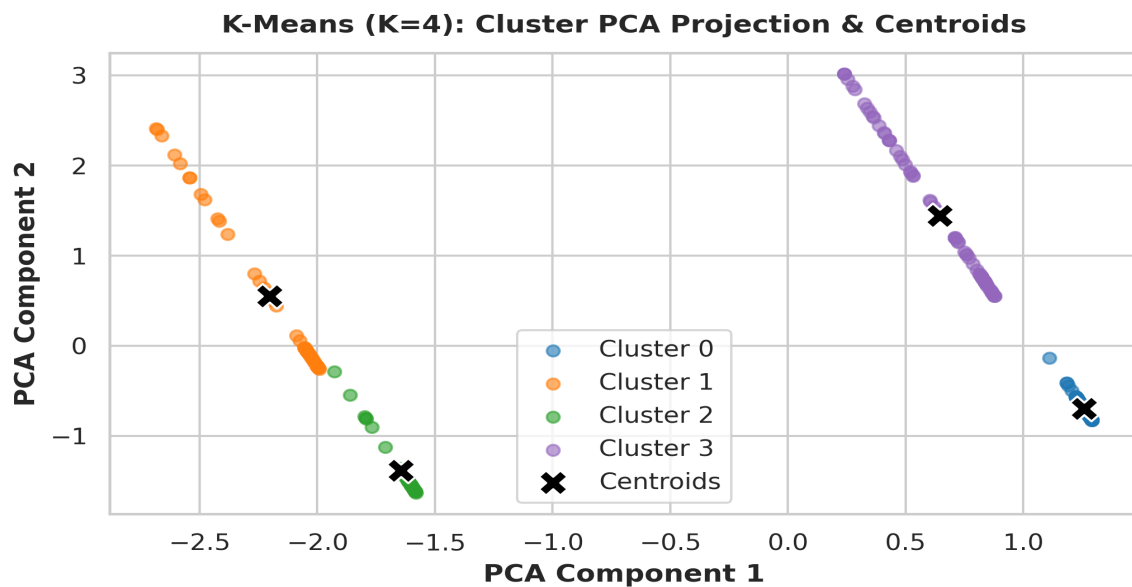
- Model Rationale:** K-Means partitions payment methods into K distinct operational clusters.
- Evaluation Metrics Achieved:**
- **Optimal K:** 4 Clusters
 - **Silhouette Score:** 0.7013 (High cluster cohesion & separation)

Visualizations for K-Means Clustering:



WHAT Chart 5 shows: Elbow Curve (Inertia) & Silhouette Score across K=2..6.

WHY visualize this? Mathematically proves K=4 is the optimal cluster number.



WHAT Chart 6 shows: PCA projection of 4 distinct clusters with marked black centroid 'X' markers.

WHY visualize this? Demonstrates clear operational tier segmentation.

6. Presentation Script & Viva Talking Points for Ma'am

Use these exact line-by-line talking points when presenting Phase 4 to your professor:

- **Introduction:** 'Respected Ma'am, in Phase 4 of our project, we built a comprehensive Machine Learning pipeline consisting of Feature Engineering, Classification, Anomaly Detection, and Clustering.'
- **Feature Engineering:** 'We constructed 6 binary, numeric, and categorical features from our Silver Lakehouse dataset, including upi_present, crypto_present, bank_account_present, amount, and site_name.'
- **Random Forest Classification:** 'Our Random Forest classifier achieved 72.00% Accuracy and 69.92% F1 Score in categorizing payment methods based on feature attributes.'

- **Anomaly Detection:** 'We trained an Isolation Forest model which successfully detected 13 outlier payment configurations (5% contamination rate).'
- **Clustering & Evaluation:** 'Finally, we implemented K-Means Clustering (K=4), achieving a Silhouette Score of 0.7013, proving clear operational tier segmentation.'